

TakeHomeQuiz4.7

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tinytex)
```

4.7

```
UNLifeExpectancy <- read.csv(file.path(str_remove(getwd(), "/Class Notes"), "/Book Data/UNLifeExpectancy.csv"))
```

```
UNLifeExpectancy <- UNLifeExpectancy %>%
  mutate(LNHEALTH = log(PRIVATEHEALTH))
```

a.)

$$H_0 : \beta_{PUBLICEDUCATION} + \beta_{LNHEALTH} = 0$$

$$H_A : \beta_{PUBLICEDUCATION} + \beta_{LNHEALTH} \neq 0$$

If the partial F-test p-value is less than $\alpha = 0.05$, then we reject the null hypothesis.

```
UNLeregaF <- lm(LIFEEXP~FERTILITY+PUBLICEDUCATION+LNHEALTH, data=UNLifeExpectancy)
UNLeregaR <- lm(LIFEEXP~FERTILITY, data=UNLifeExpectancy)
SSEaF <- sum(residuals(UNLeregaF)^2)
SSEaR <- sum(residuals(UNLeregaR)^2)
s2aF <- SSEaF/df.residual(UNLeregaF)
FRatioA <- ((SSEaR-SSEaF)/(df.residual(UNLeregaR)-df.residual(UNLeregaF)))/(SSEaF/df.residual(UNLeregaF))
FRatioA
```

```
## [1] 0.9472476
```

```
1-pf(FRatioA, df.residual(UNLeregaR)-df.residual(UNLeregaF), df.residual(UNLeregaF))
```

```
## [1] 0.5521699
```

Since the p-value from the partial F-test of 0.55 is greater than $\alpha = 0.05$. We fail to reject the null hypothesis. This means that there is no statistically significant evidence that PUBLICEDUCATION and LNHEALTH are jointly significant in determining the life expectancy of a country.

```
UNLifeExpectancy %>%  
  ggplot(aes(x=REGION, y=LIFEEXP, group=REGION)) +  
  geom_boxplot() +  
  theme_gray() +  
  labs(title="Boxplot of Region to Life Expectancy",  
        x="Region",  
        y="Life Expectancy")
```



b.)

From the box-plot above, we can see that different regions have their own distribution of life expectancy in their respective countries. However, Region 6 and 8 have medians very different from the other regions. Where their medians are not within the inter-quartile range of the other regions. In other words, it would be safe to say that the region specific means are not similar to each other.

c.)

$$H_0 : \beta_{REGIONi} = 0 \text{ for all } i \text{ in } REGION$$

$$H_A : \text{At least one } \beta_{REGIONi} \neq 0$$

If the F-test p-value is less than $\alpha = 0.05$, then we reject the null hypothesis.

```
UNLeregC <- lm(LIFEEXP~as.factor(REGION), data=UNLifeExpectancy)
UNLeregCN <- lm(LIFEEXP~0, data=UNLifeExpectancy)
anova(UNLeregCN, UNLeregC)
```

```
## Analysis of Variance Table
##
## Model 1: LIFEEXP ~ 0
## Model 2: LIFEEXP ~ as.factor(REGION)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      185 854321
## 2      177  7649  8   846673 2449.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since the F-test p-value of essentially 0 is less than $\alpha = 0.05$. We reject the null hypothesis. This means there is statistically significant evidence that at least one region is significant in computing a country's life expectancy.

```
UNLeregD <- lm(LIFEEXP~FERTILITY+PUBLICEDUCATION+LNHEALTH+as.factor(REGION), data=UNLifeExpectancy)
```

d.)

```
summary(UNLeregD)
```

i.)

```
##
## Call:
## lm(formula = LIFEEXP ~ FERTILITY + PUBLICEDUCATION + LNHEALTH +
##     as.factor(REGION), data = UNLifeExpectancy)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1256  -2.3435  -0.0331   2.6997  15.4999
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      83.3971     2.4313  34.301 < 2e-16 ***
## FERTILITY        -2.7559     0.4546  -6.062 1.16e-08 ***
## PUBLICEDUCATION  -0.4333     0.2111  -2.053 0.04197 *
## LNHEALTH         -0.7939     0.7408  -1.072 0.28575
## as.factor(REGION)2 -3.9716     1.8818  -2.111 0.03658 *
## as.factor(REGION)3 -0.8854     1.7894  -0.495 0.62151
## as.factor(REGION)4 -7.1598     2.3946  -2.990 0.00329 **
```

```
## as.factor(REGION)5 -4.0960      3.2581 -1.257  0.21077
## as.factor(REGION)6 -14.3567     1.8663 -7.693 2.26e-12 ***
## as.factor(REGION)7 -4.8391      1.9036 -2.542  0.01210 *
## as.factor(REGION)8  3.8319      1.9069  2.009  0.04639 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.062 on 141 degrees of freedom
## (33 observations deleted due to missingness)
## Multiple R-squared:  0.8038, Adjusted R-squared:  0.7899
## F-statistic: 57.76 on 10 and 141 DF, p-value: < 2.2e-16
```

```
CMatdi1 <- matrix(c(1, 2, 5, 1, rep(0,times=7)), nrow=1)
CMatdi2 <- matrix(c(1, 2, 5, 1, rep(0, times = 4), 1, 0, 0), nrow=1)
betaHatd <- coef(UNLEregd)
CMatdi1 %*% betaHatd
```

```
##           [,1]
## [1,] 74.92505
```

```
CMatdi2 %*% betaHatd
```

```
##           [,1]
## [1,] 60.56835
```

The estimated life expectancy for an Arab state with a fertility rate of 2%, public education spending at 5%, and a ln(Health Expenditure) of 1 is ~75 years.

The estimated life expectancy for a sub-Saharan African country with a fertility rate of 2%, public education spending at 5%, and a ln(Health Expenditure) of 1 is ~61 years.

```
UNLEregdii <- lm(LIFEEXP~0+PUBLICEDUCATION+LNHEALTH+as.factor(REGION), data=UNLifeExpectancy)
summary(UNLEregdii)
```

ii.)

```
##
## Call:
## lm(formula = LIFEEXP ~ 0 + PUBLICEDUCATION + LNHEALTH + as.factor(REGION),
##     data = UNLifeExpectancy)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.0828  -2.3697   0.1489   2.4736  19.9318
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## PUBLICEDUCATION    -0.2833     0.2377  -1.192   0.235
## LNHEALTH           -0.8938     0.8480  -1.054   0.294
## as.factor(REGION)1  73.7945     2.1107  34.963 <2e-16 ***
```

```
## as.factor(REGION)2 70.6976      1.8124 39.009 <2e-16 ***
## as.factor(REGION)3 75.0138      1.8295 41.003 <2e-16 ***
## as.factor(REGION)4 67.1137      2.5717 26.097 <2e-16 ***
## as.factor(REGION)5 69.7726      3.7261 18.725 <2e-16 ***
## as.factor(REGION)6 54.6604      1.6114 33.922 <2e-16 ***
## as.factor(REGION)7 73.6802      1.7311 42.562 <2e-16 ***
## as.factor(REGION)8 81.9293      1.9943 41.082 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.836 on 146 degrees of freedom
## (29 observations deleted due to missingness)
## Multiple R-squared:  0.9932, Adjusted R-squared:  0.9928
## F-statistic: 2137 on 10 and 146 DF, p-value: < 2.2e-16
```

```
CMatdii <- matrix(c(0, 0, 1, rep(0, times=4), -1, 0, 0), nrow=10)
Xdii <- model.matrix(UNLEregdii)
s2d <- sum(residuals(UNLEregdii)^2)/df.residual(UNLEregdii)
se <- sqrt(s2d*t(CMatdii)%*%solve(t(Xdii)%*%Xdii)%*%CMatdii)

t(CMatdii)%*%coef(UNLEregdii)+c(-1,1)*qt(0.975, df.residual(UNLEregdii))*se
```

```
## Warning in c(-1, 1) * qt(0.975, df.residual(UNLEregdii)) * se: Recycling array of length 1 in vector
## Use c() or as.vector() instead.
```

```
## Warning in t(CMatdii) %*% coef(UNLEregdii) + c(-1, 1) * qt(0.975, df.residual(UNLEregdii)) * : Recyc
## Use c() or as.vector() instead.
```

```
## [1] 15.40445 22.86386
```

We are 95% confident that the difference in life expectancy between an Arab State and a Sub-Saharan African Country can be found between 15.4 years and 22.9 years.

```
CMatdiii <- c(rep(0, times=8), 1, 0, -1)
abs(CMatdiii %*% coef(UNLEregd))
```

iii.)

```
##           [,1]
## [1,] 18.18863
```

The point estimate in the difference between a Sub-Saharan African Country and a high-income OECD Country is ~18.2 years.